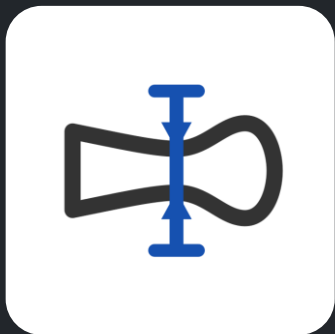




Analyze footprint

Measure any sidecut: dimensions, waist, taper, tip and tail length, average and natural radius, widest and inflection points -- and publish them as variables.

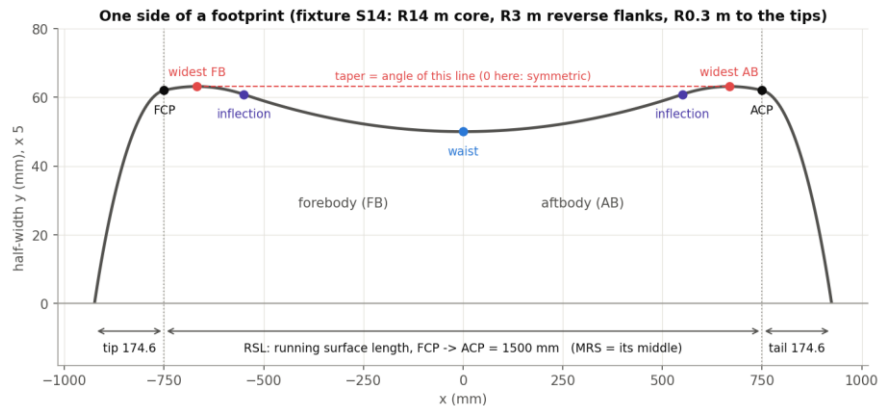


What it does

Comparing sidecuts needs the same numbers, measured the same way: tip-waist-tail, taper, radius. Analyze footprint reads them off any outline -- designed, scaled or imported -- and publishes them for later features.

Use it when

- Check a sidecut against its targets
- Compare skis in a family
- Drive later features from measured values

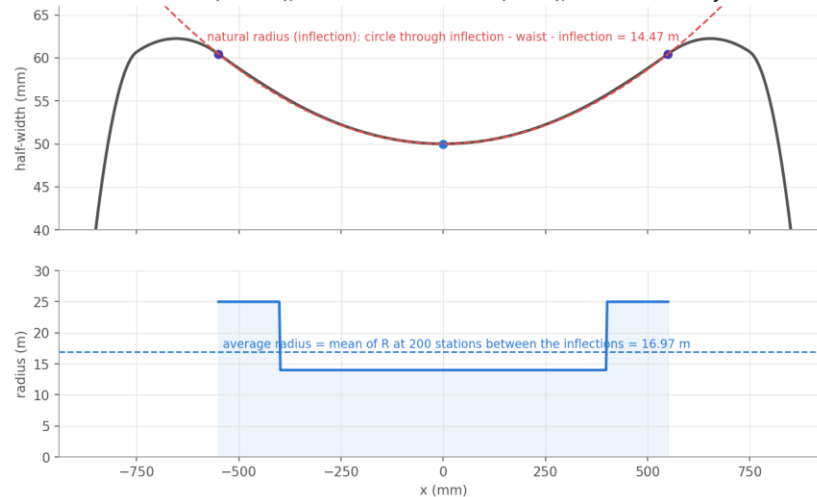


The points it finds: contacts, widest, inflections, waist.



Average vs natural radius

Fixture FX2: R14 m core (x +-400), R25 m to the inflections (+-550), reverse flanks beyond the inflections



Average

Mean of R at 200 X stations between the inflections -- independent of how the edges are split.

Natural

One circle through inflection - waist - inflection (or widest - waist - widest).

They differ

FX2: 16.97 m average vs 14.47 m natural.
Both are right; they answer different questions.



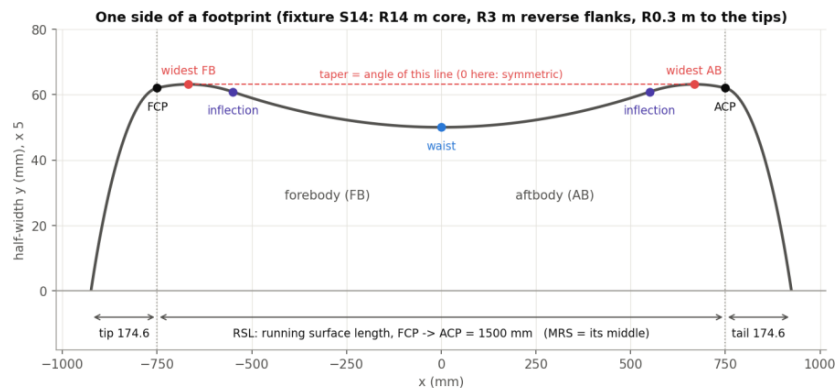
The dialog

IMAGE MISSING:
docs/decks/analyze_footprint/shots/dialog.png

#	Parameter	What it means
1	Footprint edges	The sidecut (+Y side).
2	RSL line	A straight edge FCP to ACP.
3	FCP	Plane, vertex or mate connector; the nearest point on the RSL line is used.
4	ACP	As FCP.
5	Sketch key points	Draw the waist, widest and inflection lines.
6	Footprint Data	Read-only: tip-waist-tail, waist width and X, taper, tip and tail length.
7	Radius calculations	Average, natural (widest), natural (inflection).
8	Widest & Inflection points	Distances from each contact; inflection to widest.
9	Recalculate	Button: refresh the panel.



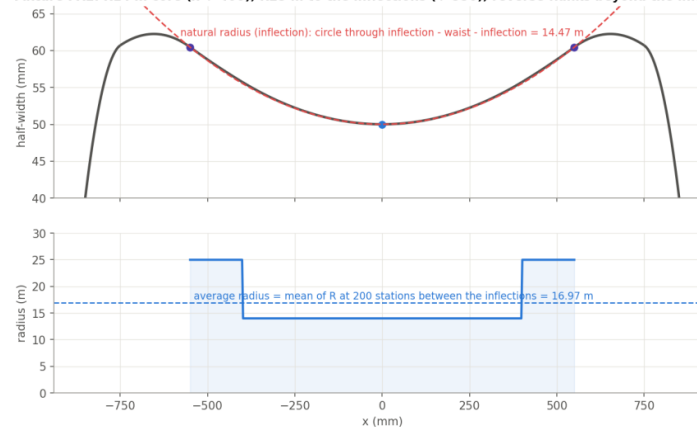
Examples (Footprint tests)



AF1: S14

Inflections ± 550 , average 14.00 m, natural (inflection) 14.00 m, natural (widest) 17.00 m, taper 0.

Fixture FX2: R14 m core (x ± 400), R25 m to the inflections (± 550), reverse flanks beyond the inflections



AF2 / AF3: FX2 in 7 or 13 edges

Same result either way: average 16.97 m, natural 14.47 m.



Outputs and messages

Published for Extract variables

Key	What it holds
tipWidth, waistWidth, tailWidth	Full widths at the widest points and the waist.
waistX, tipLength, tailLength	World X of the waist; contact to tip / tail end.
taperAngle	Widest to widest; positive when the tip is wider.
sidecutRadius	The average radius.
output	The key-point sketch (when on).

Messages

	Message / what to do
Info	Tip-waist-tail ... mm average radius ... natural radius (inflection) ... Every regeneration, always current.



Tips and limits



Panel goes stale

The dialog panel updates only when the feature is edited or Recalculate is pressed; the info line and variables are always current.



Any orientation along X

Tip at +X and off-centre skis measure correctly (AF4, AF5); raw sketch arcs too (AF7).



Where it lives

Document	footprint (52724f3a857fa52d3ecceb77)
Tab	analyzeFootprint (+ fpt_analyze, predicates)
Tests	Footprint tests studio (build_ / check_footprint_tests.py): AF1-AF7
Explainer	docs/explainers/footprint/footprint_explained.md, section 1.3, 2.2
Related	Integrate footprint, Analyze footprint, Scale Footprint, Arc fit, Generate Footprint Points